ECOLOGY, ECONOMY AND REDEMPTION AS DYNAMIC: THE CONTRIBUTIONS OF JANE JACOBS AND BERNARD LONERGAN

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Abstract

Bernard Lonergan, S.J. and Jane Jacobs have devoted much of their intellectual careers to thinking out the dynamic natural-human environment. Lonergan and Jacobs worked in very different lines of research—systematic theology and urban economics, respectively. Despite predictable differences in their thought, there are also remarkable commonalities in their analyses. Both thinkers have argued that the same dynamic principles that govern the functioning of natural ecologies are also to be found when human social and economic systems function well, but are absent when human systems go wrong. Both have argued that the violation of principles that pertain to natural ecologies is destructive not only of the natural environment, but of communal and economic well-being as well.

Jacobs came to prominence with the 1961 publication of her classic, *The Death and Life of Great American Cities*. She has since gone on to extend her analysis to the unique characteristics of urban economics in several books and articles. In her most recent book *The Nature of Economies* (2000), Jacobs draws the results of her previous work on urban economic patterns into a synthesis with recent insights into biological systems. She argues that exactly the same principles (or "processes" as she prefers to call them) that sustain vital, evolving natural ecologies also underpin robust and dynamic economies. Where Jacobs's work gives a richly detailed account of the processes shared alike by natural and human systems, Lonergan developed a parallel, integral account of natural processes, human social and economic organization, and the "economy of salvation." In his classic work, *Insight*, Lonergan argues that the dynamics of human innovations and self-correction correspond in striking ways to the emergence, growth, development, and decline in the natural order. Unlike natural ecologies, however, the possibilities of genuine social and economic development are distorted, Lonergan argues, by the forces of "bias." In his role of theologian, Lonergan goes on to explore how divine grace heals the distorted dynamics of natural and human ecologies.

Keywords: Bernard Lonergan, Jane Jacobs, environment, ecology, urban economics

Introduction

> *Economic development isn't a matter of imitating nature. Rather, economic development is a matter of using the same principles that the rest of nature uses.* (Jane Jacobs 2000: 31)

It has been remarked that in "the past, ecologists often assumed a dichotomy between a pristine, stable nature and disruptive human activity."[1] Perhaps, then, it is best to begin with a statement of fact, at risk of stating what may already be obvious to the reader: nature is neither stable nor pristine. Before a single hominid walked the surface of this wonderful planet, nature was constantly being disrupted by processes of evolution, development, and eradication. Nor did nature desist from such processes once humans appeared, ceding all future evolutionary change to the initiatives of humans. In no way do I mean this observation to provide any sort of rationalization for the kinds of abuses of natural environments that we have witnessed over the past two centuries. Rather I wish to emphasize that if we are to be properly critical of human deeds and their impact upon the natural environment, we must carefully and critically understand natural developmental processes and understand the kinds of human developments that are compatible and that are incompatible with natural development.

In this paper I intend to explore the thought of two authors who have devoted much of their intellectual careers to thinking out the dynamic relationships between the natural and human environments, Bernard Lonergan, S.J. and Jane Jacobs. Lonergan and Jacobs worked along very different lines of research—systematic theology and urban economics, respectively. Despite predictable differences in their thought, there are also remarkable commonalities in their analyses. Both thinkers have argued that the same dynamic principles that govern the functioning of natural ecologies are also to be found when human social and economic systems function well, but are absent when human systems become corrupted: "economic development is a matter of using the same principles that the rest of nature uses" (Jacobs 2000: 31). Both have argued that the violation of principles that underpin natural ecologies is destructive not only of natural environments, but also of human economic and communal well-being. In this article I hope to make available the seminal ideas of these two thinkers, and to point to lessons that may be drawn from their analyses in framing policy issues.

Bernard Lonergan: The Emergence of Nature, the Human Order, and Grace

Making ethical decisions, informed by faith, about human economic initiatives that impact natural environments.

Educated in Catholic scholastic philosophy and theology of the 1920's and 30's, the young Bernard Lonergan soon recognized that the two great challenges confronting Catholic theology were the intellectual transformations effected by modern natural science and by history. Already in his student days he was energetically at work on forging a new framework that would integrate and reconcile the account of natural processes that emerge from modern scientific research with human moral aspirations and the workings of divine grace.

Those efforts reached a mature culmination in his classic work, *Insight: A Study of Human Understanding*. There he gradually develops his account of the dynamic processes of the created order, characterized by what he called "emergent probability." While Lonergan's account is evolutionary in the broad sense, he carefully distanced himself from Darwin's version of evolution and the reductionistic, materialistic biases that characterize much of neo-Darwinian thought. In Lonergan's view, materialism assumes that ultimate reality is known through sensation, especially through sight and touch (i.e., the "real" as what resists when our bodies come into contact with it). He referred to this assumption as the "already out there now" view of reality. By way of contrast Lonergan argues that what is known through sensation is only a component of reality. Rather, it is intelligibility (what is added over and above sense knowledge by human insight and judgment) that is the real heart of the natural reality. Natural science, he argues, is fundamentally concerned with discovery of the intelligible relationships and orders that make up the natural world. Where a materialistic worldview regards something like physical impact as the ultimate explanatory cause, Lonergan argues that it is, rather, intelligible relatedness that explains why things are as they are and why they behave as they do. Nowhere is this more evident than in Lonergan's account of "emergence." In the following paragraphs I offer a brief sketch of Lonergan's account of "emergent probability."

Schemes of Recurrence

Lonergan remarks that the focus of Darwin's evolutionary accounts was the gradual accumulation of small "sensible qualities," (1992: 290) that is to say, observable and describable phenotypic characters. By way of contrast, the focus of Lonergan's account is what he calls "schemes of recurrence." The small variations of classical Darwinism do not merely pile up. Rather, scientifically they must be understood in their intelligible relationships to the internal and external functioning of the organism and its environment:

> the concrete living of any plant or animal may be regarded as a set of . . . recurrent operations . . . Within such schemes [of recurrence] the plant or animal is only a component. The whole schematic circle of [operations] does not occur [solely] within the living thing, but goes beyond it into the environment (1992: 156).

By "scheme of recurrence" Lonergan means a series of events (or "operations") that are intelligibly linked together by natural laws of physics, chemistry, biology, etc. Schemes of recurrence can be "represented by the series of conditionals, If A occurs, B occurs; if B occurs, C occurs; if C occurs, . . . A will recur" (1992: 141). The intelligible *connection* between the occurrence of A and B, between B and C, etc. is determined by some law of physics, chemistry, biology, etc. Simple examples of schemes of recurrence include the hydrogen-helium fusion cycles in the interiors of stars, the Krebs cycles in cells that continually regenerate energetic ATP from depleted ADP, and the mutual regeneration of atmospheric CO_2 and O_2 by animals and plants. Lonergan goes on to note that schemes of recurrence are usually far more complex, involving intricate sub-loops and alternative pathways.

Lonergan's focus on schemes of recurrence places emphasis on two important facts. First, the evolving unit is not a random material variation of the organism (e.g., a longer limb) but a way of living (i.e., an intelligible functioning). Second, the environment is not merely a passive stage onto which an individual organism with a variant feature is placed. The schemes of the environment are always already implicated in every organism's functioning.

Emergence

Emergence has always been a problem for hard-core materialism, which tends to regard underlying matter (elementary particles) as ultimately unchanging. Although the rearrangements of matter change, no change occurs in ultimate reality, according to materialism. In its tacit denial of the reality of emergence, this materialist philosophy tends to fly in the face of common sense and religious belief. Interestingly, Lonergan argues that materialism is a *philosophical* position that is also incompatible with *scientific* study. Science, he argues, seeks correct understanding of how events are intelligibly connected within schemes of recurrence. When new schemes begin to function, *really new* intelligibilities emerge, and it is the task of science to correctly understand those newly emergent intelligibilities.

What is impressive about Lonergan's account of emergence is how he can avoid materialism without needing to invoke any sort of vitalistic force or *elán vital*. Instead, he draws attention to an obvious but commonly overlooked feature of laws of science: their conditionality (1992: 70). Laws of science are extremely general, but for that very reason they are also are extremely indeterminate. For example, Newton's three laws of motion and his law of gravity are extremely general; they apply to any massive object. Yet from those laws *alone* it is impossible to derive any specific, concrete path of motion. A determinate path of motion may be deduced *only* once *specific* initial conditions are stipulated. For example, if one stipulates or observes two bodies with very special combinations of mass, position and velocity, then a particular elliptical path can be deduced; but for a different combination of conditions, the path will be hyperbolic. As another example, the laws of chemistry allow that if octane is combined with oxygen, then carbon dioxide and water will be produced ($2\ C_8H_{18} + 34\ O_2 \rightarrow 8\ CO_2 + 18\ H_2O$). However, this chemical transformation will occur as *complete* oxidation, only under highly specific conditions of pressure, temperature, concentration, catalysis, etc. Under different conditions, carbon monoxide, carbon particles and smaller hydrocarbon chains will be produced. Both sets of outcomes are completely intelligible in terms of the very same laws of chemistry. To put the matter bluntly: the laws of science in and of themselves determine nothing. It is only the laws plus specified conditions that determine concrete events (1992: 70-71).

Lonergan used this feature, the conditionality of scientific laws, as the foundation of his account of emergence of schemes of recurrence. In the scheme of recurrence, "If A occurs, B will occur; if B occurs . . ." the occurrence of A is a condition for the occurrence of B. That does not mean, however, that A is the one and only condition for the occurrence of B. In general, we might think of A as the last condition to fall into place so that B occurs. The only thing that singles A out for special consideration from all the other conditions requisite for B is that A just happens to be an event that sets off a self-conditioning scheme that eventually leads back to the recurrence of A. In other words, *if* all of the other appropriate conditions happen to be already fulfilled (i.e., "other things being equal"), then the occurrence of A will result in the occurrence of B and "if B occurs, C will occur; if C occurs . . . A will recur."

For Lonergan, then, there is such a thing as real emergence. Schemes of recurrence emerge. Schemes of recurrence are not merely spatial aggregations of material particles or random variations. They are really distinct, novel, intelligible functionings. They emerge wholly in accordance with laws of science. No vitalistic force is needed to produce them or breathe life into them. Yet schemes emerge only when the appropriate prior conditions happen to be fulfilled. Lonergan goes on to explain how the assembly of appropriate prior conditions occurs randomly, and hence that there are "probabilities of emergence and probabilities of survival" that pertain to this field of environmental conditions. Indeed he argues that such probabilities undergo dramatic jumps, but that is tangential to the purposes of the present paper.[2]

Conditioned Sequences of Emerging Orders and Development

Lonergan's explanation of emergence is rolled into his new kind of evolutionary account, "emergent probability," by noting that certain kinds of schemes of recurrence themselves form the prior conditions for other subsequent schemes (1992: 145). He writes for example that a biological species "is an intelligible solution to a problem of living in a given environment"; that "later species are solutions that . . . rise upon previous solutions"; and that "a solution is the sort of thing that human insight hits upon" (1992: 290). He goes on to explore the various features of this approach to emergence and evolution. He argues that emergent probability has the same scope of explanatory potential as Darwinism, but without the latter's reductionistic tendencies (1992: 145-151, 155-57). Later he presents a stronger argument that emergent probability is indeed the intelligible order of the natural world.[3]

Lonergan draws attention to a perennial tension that comes to light from this account of emergent probability, a tension that is especially pertinent to questions about the proper relationships between human development and the natural environment:

> [N]o less than stability, the possibility of development must be considered. Unfortunately, these two can conflict. Schemes with high probabilities of survival tend to imprison materials in their own routines. They provide a highly stable basis for later schemes, but they also

> tend to prevent later schemes from emerging. A solution to this problem would be for the earlier conditioning schemes to have a high probability of emergence but a low probability of survival (1992: 146-47).

Consider for example the enormous numbers of offspring produced by certain species (e.g., salmon), where on average only two survive to reproductive maturity, the remainder serving as food for other living organisms. Simply put, the intelligibility of strictly natural emergent probability is compatible with the termination and even extinction of particular schemes of recurrence. Lonergan's way of situating the "conflict" of development within the larger context of evolutionary emergent probability raises a significant issue. On the one hand, emergence depends on the continued functioning of prior recurrent schemes that constitute ecosystems. If these are violently destroyed, both they and the subsequent emergent forms that depend upon them are lost. On the other hand, development cannot proceed without the transformation of prior schemes by later ones. In nature, emergent probability "respects" its underlying conditions, and yet it does not leave prior schemes untouched. Both Lonergan and Jane Jacobs seek to comprehend how human development can profit by learning to respect this delicate interplay.[4]

Human Schemes of Cooperation

Lonergan goes on to show the pertinence of emergent probability to the realm of human affairs. He points out that the human world is permeated by schemes of recurrence.

> As in the fields of physics, chemistry, and biology, so in the fields of human events and relationships there are . . . schemes of recurrence. For the advent of [humanity] does not abrogate the rule of emergent probability. Human actions are recurrent; their recurrence is regular . . . [but] their functioning is [conditioned,] not inevitable (1992: 234-35).

In his account, human social and economic schemes operate, not with the blind laws of natural selection, but through the conscious, self-correcting activity of human inquiry and insight. Human schemes consist of intelligible patterns of relationships that "condition the fulfilment of each man's desires by his contributions to the desires of others" (1992: 239). For example, every commercial enterprise is a scheme of recurrence of human actions that involves recurrent transactions among suppliers, workers, buyers and recurrent patterns of payments that condition their functioning. Every family involves recurrent schemes that not only continually takes in sustenance from the economy and regularly disposes of waste products, but also regularly distributes mutual acts of love, reassurance, support, discipline, and education amongst its members.

What radically distinguishes human schemes of recurrence from natural ones is that their emergence and survival depend upon acts of human intelligence and choice. Human "practical intelligence devises arrangements for human living" (1992: 239). These arrangements are largely patterns of cooperation that depend upon understanding "what one can expect" of the other person (1992: 248). Human practical intelligence (or "common sense") is the accumulation of innumerable such insights that make possible the full participation in human economic, social and political institutions.

Human insights not only maintain the schemes of human living; they also constantly transform and bring about new schemes. Prior schemes "set problems calling for" insights into inventions, insights into how to organize to distribute the fruits of production ever better, insights into how to reach group agreements and decisions ever more fairly and effectively (1992: 233-34). The ongoing development and emergence of human schemes follows what Lonergan called a "self-correcting" process: existing human scheme → questions about how to do thing better → insights into improvements → actions that modify the schemes → further questions and insights, and so on.

> The creative task is to find answers. It is a matter of insight, not of one insight but of many, not of isolated insights but of insights that coalesce, that complement and correct one another, that influence policies and programs, that reveal their short-comings in their concrete results, that give rise to further correcting insights (1999a: 100).

This is a most interesting aspect of Lonergan's understanding of creativity. Lonergan's is not a Romantic notion of creativity that spurns critical intelligence and rationality. Creativity in Lonergan's sense includes and depends upon intelligent criticism. In principle, intelligent self-correction of human patterns of cooperation has the potential to respond intelligently to every difficulty. It can bring about not only technological innovations but also equitable economic distribution and political justice as well. Asking and answering numerous further questions is required to take into account both the immediate and long-term consequences upon intricate networks of recurrent schemes that characterize natural and human ecologies alike. As Lonergan notes, fully intelligent and ethical choices "cannot consistently" undertake initiatives that destroy their underlying conditions, including natural ecological conditions (1992: 629).

The potential of human emergent probability for ongoing improvement is underpinned by what Lonergan calls the "pure *unrestricted* desire" to understand, know and realize the good. In principle human beings have an unlimited capacity to ask every sort of question and face every sort of problem posed by any scheme of human cooperative living, including questions pertaining to proper care and protection of the natural environment.

Bias and Destruction

To say that human beings have an unlimited capacity to inquire intelligently into every sort of problem posed by their organized efforts does not, of course, mean that human beings actually do so. Unlike natural ecologies, innovations in human social and economic arrangements are all too frequently implemented without the fullness of intelligent self-correction. Real self-correction can occur only when the full complement of further pertinent questions and problems are taken into account and answered by with creative solutions. The possibilities of genuine social and economic self-correction are cut short, Lonergan argues, by the forces of what he terms "bias." His views therefore resonate with Jane Jacobs', when she complains that, "we no longer care" to *understand* "how things really do work, but only what kind of quick easy outer impression they give" (1993: 11). When biases interfere with the full development of intelligence, both the well-functioning of human systems as well as their underlying natural infrastructure are imperilled.

Lonergan's account of emergent probability in the human order incorporates the fact of human failures to consider questions raised by their endeavours, failures to seek answers even to all the questions they do raise, and refusals to act according to what they come to understand as the best courses of action. He identifies four fundamental forms of bias that distort human collaborative efforts into dysfunctional constellations: psychological aberrations ("dramatic bias"), individual selfishness ("individual bias"), ethnic, racial, class and gender discrimination ("group bias"), and the narrow-minded disregard for non-immediate consequences, such as long-term environmental impacts ("general bias"). Instances of bias are legion.[5] They all operate by ignoring the reflective processes of asking and answering all the questions that are raised by complex situations. According to Lonergan, biased courses of action that evade intelligent self-correction initiate downward spirals of decline, degradation and destruction not only of natural but also of cultural environments. Biases and decline have their own "logic"—the logic of vicious cycles that lead to great destruction, unless something acts to reverse their downward trends (1992: 214-23, 242-63).

Grace and Redemption

In using the term "bias" Lonergan characterizes the accumulating devastation in terms of its opposition to the self-correcting potential of intelligence, inquiry, and insight. But as a Christian theologian, Lonergan was clear that the same pattern of decline is to be regarded as a pattern of sin from the viewpoint of its opposition to God. Lonergan is in fundamental agreement with St. Augustine's characterization: "evil is nothing but the removal of good until finally no good remains."[6] As a Christian theologian, he affirmed that the reversal of sin and its devastating social consequences is accomplished by God's grace. In fact his earliest research was on the development of Aquinas's theory of grace (2000).

What is distinctive in Lonergan's own treatment of grace and redemption is his way of situating them in relation to emergent probability. In *Insight*, he raises the question of God's solution to the problem of sin, evil, and social decline. He argues that the solution requires the "supernatural" virtues of "faith, hope and love" (1992: 718-25, 741). There he explores how these virtues cooperate with redemption (the economy of salvation) as it occurs within this universe of emergent probability—"When in the fullness of time" the Redeemer came, as Christian theology has put it.

Soon after *Insight*, however, Lonergan began to speak about the relationship between emergent world process and redemption more broadly as involving "three dynamics" of creativity and progress (intelligent self-correction), decline and degradation (bias and sin), and redemption and recovery through the healing that takes place in all religious love (1993, 1999a). Religious love, according to Lonergan's later view, is a constant in human affairs. Love heals hatred and bias. Love offsets the corrosive effects of stupidity and wickedness. Opposite to religious love, there is a hostility toward and hatred of nature embedded in the seminal works of some founders of modernity like Machiavelli and Bacon. There is also misanthropic hatred to be found in certain strains of environmental activism. Yet religious love is love of God, and to love God unconditionally is to love everything God loves—all of natural and human creation. Grace is religious love, and it sets about undoing hatred. Loving grace makes possible healing, discerning, and creatively intelligent responses to degenerating situations.

As a Christian theologian Lonergan identifies the unconditional love found in *all* religions with the mission of the Holy Spirit, "God's love flooding our hearts through the Holy Spirit given to us (Rom 5, 5)" (1972: 105). This dynamic of redemption, Lonergan claims, suffuses all human history and is present through all human affairs, just as are intelligent creativity and biased degradation. In Christian belief, the mission of the Holy Spirit reaches full efficacy through the mission of the Son, who inaugurates in emergent probability God-authored schemes testifying to God's undying redemptive love.

Reductionistic forms of Darwinism tend not only to ignore the role of God as Creator, but also to eliminate theological considerations of grace as superfluous and even as dangerous distractions from urgent mundane matters. Lonergan, to the contrary, integrates God's grace with the evolving character of the natural and human worlds and the challenges they pose.

Jane Jacobs: Across the Nature/Human Divide

Jane Jacobs came to prominence when in the early 1960's she spearheaded a community organizing effort to block construction of the Lower Manhattan Expressway, handing New York City's planning czar, Robert Moses, his first ever defeat. During that same period she also wrote and published her classic, *The Death and Life of Great American Cities*, which has become mandatory reading for urbanologists. Her book presented a powerful critique of the then prevailing theories of urban development that underpinned the redevelopment projects of Robert Moses's generation. Just as importantly, Jacobs also offered a nuanced alternative theory of her own, the principles of which have received wide reception and verification.

Jacobs has since gone on to extend her original line of investigation in several additional books and numerous articles. In these works she presents her analyses of the ways urban economies function and become dysfunctional. "The basic idea," she writes, is "to try to begin understanding the *intricate* social and economic order underlying the seeming disorder of cities" (1993: 21-22). In her most recent book *The Nature of Economies* (2000), Jacobs draws the results of her previous work on urban economic patterns into a synthesis with recent researches into biological and dynamic systems (sometimes called "chaos" or "complexity" theory). She argues that exclusive reliance on prevailing economic assumptions—such as marketplace "laws" regarding maximization of profit—are both too abstract and too inadequate. The same assumptions that have in many ways led to destruction of natural ecosystems, she claims, also lead to destruction of economic vitality as well. She further argues that exactly the same principles (or processes as she prefers to call them) that sustain vital, evolving natural ecologies also underpin robust and dynamic economies. According to Jacobs, both healthy biological systems and healthy economies have four common characteristics: (1) development, (2) expansion, (3) self-maintenance through "self-refueling," and (4) evading collapse. In this section I will explain her analyses of these four fundamental and interconnected characteristics, and how both ecologies and economies remain vital and sustainable only when all four processes are present and allowed to operate.

Development

> *Where do new things come from? . . . An animal, a plant, a [river] delta, a legal code, an improved shoe sole—they all depend on the same underlying process for development* (2000: 15-16).

Jacobs initially characterizes all kinds of developments as the interplay of two principles: differentiations emerging from within generalities; and differentiations becoming new generalities from which further differentiations then emerge (16-17). (This closely parallels Lonergan's account of how schemes of recurrence emerge from prior conditioning schemes). By her imaginative and strategic use of illustrations, Jacobs makes a persuasive case that both natural and human developments all follow the same patterns. For example: (a) An originally generic, undifferentiated cosmic cloud became differentiated into a star (our sun) and nine very different sorts of planetary climate systems. One of those (our earthly system) became in turn the new "generality" within which diverse ecosystems and life forms differentiated. (b) In mammalian evolution, successive sequences of differentiations of vertebrate forelimbs have produced hoofs, paws, hands, flippers, and wings. (c) In embryogenesis, the generically indistinguishable cells of a zygote differentiate into ectodermic, mesodermic, endodermic cellular layers, which in turn become generalities that further differentiate into the multifaceted tissues and systems of an adult mammal. (d) The first crude wheel, whatever its origin, has been modified, and its modifications modified, over and again into such differentiations as rimmed spoked wheels and "rimless spoked wheels such as: water mill-wheels, windmills, fans, paddle wheels, propellers, food blenders," solid wheels like "the potter's wheel . . . circular saws, rotating dials, phonograph turntables, movie projectors," and so on (2000: 25).[7]

To the interplay of these two principles, Jacobs adds a third: all development depends upon co-development. She insists that it is a great mistake to think of development linearly, as is suggested in biology textbooks by evolutionary trees that trace lines of descent. Such "trees" are too abstract. They prescind from the real fact that concretely, evolution operates not *linearly*, but "as a *web* of interdependent co-developments" (2000: 19). In other words, any newly differentiated innovation needs a habitat, and habitats consist of intricately interconnected differentiations, each of which itself had to have been developed.[8] This is evident in the intricacies of evolved natural ecologies. It is no less evident in human economies, Jacobs argues. In this way Jacobs contextualizes and relativizes competition and survival: "Competition for feeding and breeding take place in an area. That area is a [co-developed] habitat" (21).

Expansion

> *Development is qualitative change. Expansion is quantitative change. The two are closely linked* (2000: 37).

The second component in Jacobs' analysis, expansion, begins with the question, "Why don't new developments crowd old ones out?" From this question flows her analysis of how both biological and economic environments expand. "The most amazing demonstration of expansion is the sheer volume and weight of biomass on the earth. It expanded from nothing before life began" (2000: 43). Expansion here can mean growth in spatial extent, but more to the point, it means growth in complexity. Webs of co-developments do not merely multiply instances; they multiply differences and intertwine differences together into ever more complex systems. In this way co-developments are constantly providing new interstices and niches for still further developments.[9] This is why old developments are not automatically crowded out.

Not content, however, to merely solve the crowding problem, Jacobs offers an account of how the dynamics of expansion work. Reminiscent of the work of Ilya Prigogine,[10] she explains that

> Expansion depends on capturing and using transient energy. The more different means a system possesses for recapturing, using, and passing around energy before its discharge from the system, the larger are the cumulative consequences of the energy it receives (2000: 47 & 46).

In order to illustrate her point, Jacobs contrasts desert with forest environments. Both receive comparable amounts of energy from the sun. Yet in forests:

> energy flow is anything but swift and simple, because the diverse and roundabout ways that the system's web of teeming, interdependent organisms uses energy. Once sunlight is captured, it's not only converted but repeatedly reconverted, combined and recombined, cycled and recycled . . . Energy flow through an intricate conduit of this kind . . . leaves behind, complex webs of life (2000: 46).

By way of contrast, deserts have relatively few systems for capturing sunlight and converting it into forms usable by other biological systems. The lesson that she wishes to draw from this analysis is that expansion is not primarily a function of external inputs—sunlight—but rather of internal complexity.

Jacobs then directs her analysis to the phenomena of economic growth.

> I began thinking about settlements' economies as instances of natural energy-flow [and realized] that imports came in at the receiving end of their conduits, exports left at the discharge end, and the interesting question was what went on within the conduits (2000: 53).

What "goes on within the conduit" is the complex interdependent patterns of working, producing, trading, and living that characterize each particular settlement. Here Jacobs incorporates her earlier work on the dynamics of urban economies (1969, 1985) into the more comprehending world-view linking natural and human orders. She argues that theories and policies intent upon promoting development have focussed too much upon introducing external inputs—such as large grants and loans or large scale "development projects." This excessive attention to large energy-like inputs has led to severe neglect of both the existing webs of complexity and the means needed for differentiating and diversifying patterns that already exist. The result of input theories, she argues, has been devastation both of economic and natural ecologies.

Self-maintenance through Self-refueling

> *Machines lack equipment for refueling themselves* (2000: 66).

Jacobs distinguishes dynamic natural ecologies and economic ecologies from mere machines by their capacities to "refuel" themselves. What she means by this is that part of the system's operation is devoted to obtaining the forms of "energy" usable by the system itself. "Part of the energy each takes in from outside itself is spent to capture subsequent infusions of energy, and part of that to capture more infusions, and so on, repeatedly" (2000: 65). Mere mechanical systems, by comparison, are passive with regard to energy acquisition; someone else has to turn the crank or fill the tank, so to speak.

In line with her endorsement of the value of diversity and complexity in other contexts, here, too, Jacobs insists that

> Each system has its own integrity as a discrete, tangible unit. One organism's waste is another's dinner. Self-refueling has no generalized form—only many, many specific forms (2000: 67).

Because self-refueling is so crucial to a system's integrity and survival, this assertion of uniqueness implies, therefore, that it is imperative to learn how each system effects its own form of self-refueling, and to undertake no policy that would imperil it. Once again drawing upon her earlier work, Jacobs argues that vital economic systems refuel themselves by beginning to produce locally what they had previously been importing from outside. Earlier she had devoted much attention to the dynamics of this "import replacing" process (1969, 1985); here she is integrating that earlier analysis into her broadened framework of natural/human ecology.

Jacobs explains that she deliberately coined the term, "self-refueling" in order to avoid certain kinds of moral principles associated with environmental movements that she has found inadequate:

> What about *self-relying, self-sustaining,* and *sustainable?* . . . Those expressions overlap with *self-refueling,* although we tend to give them moral overtones. For example *self-reliance* is generally taken to be so admirable that lack of it is seen as unfortunate or even bad. *Sustainable* commonly applies to the practice of drawing on renewable resources at a rate no needier or greedier than the rate at which the resources can renew themselves; the practice implies environmental morality. Self-refueling [rather] is a basic natural process . . . so fundamental to survival . . . that conceptions of whether it is a good or bad thing are pointless (2000: 67-68).

Her point is that simplistic versions of these moral principles can interfere with really understanding how refueling works—preemptively classifying certain forms as "bad" before they are properly understood in relation to their environments. This is not to say, however, that Jacobs is unconcerned with the rate at which resources can renew themselves. Quite the contrary; she explicitly acknowledges that all natural systems and human settlements require unearned "gifts" (Lonergan's earlier 'schemes') that are "inheritances from earth's past developments and expansions" in order to start their own developments (2000: 54-55; more on the theological significance of this observation later.) Her opposition is to policies based on overly simplistic moral principles that would declare certain categories of innovations in self-refueling to be "off limits" before the concrete situations are fully explored. Her counsel is to pay attention, instead, to examples from healthy economies where "chains of replacements typically start with goods and services that are easiest to replace at a specific time and in a specific place and replacements can proceed to more complex and difficult ones as . . . its refueling equipment . . . diversifies and expands" (2000: 78-79).

In dynamic ecologies self-refueling is itself a dynamic process. It is no mere matter of constantly taking in the same forms of energy in the same way year in and year out. She points out that if this were so, the system would eventually perish. The constancy of the form of its input energy cannot be guaranteed; moreover, by its own expansion, it produces ever increasing amounts of dissipated energy that begin to accumulate in its own environment. Unless the system adapts, it will cease to function.

In any healthy biological or economic ecosystem, therefore, the dynamics of self-refueling both presuppose and are presupposed by the two previously discussed dynamics of development and expansion (2000: 82). Self-refueling takes in appropriate forms of "energy" (inputs, imports). Development modifies the system's "equipment" including its mechanisms of self-refueling. Expansion converts, recycles, and recombines the energy, altering its dissipated energy (outputs, exports). Altered outputs provide new potential energy sources, which further development, expansion, and self-refueling can make use of. (This pattern is comparable to Lonergan's account of the series of later schemes that emerge from the functionings of earlier ones.)

Evading Collapse

> *All dynamic systems are in danger of succumbing to instability, which is why they need constant self-correction* (2000: 85).

Because there is no such thing as a perfect "total system," dynamic biological and economic systems characterized by development, expansion and self-refueling also need processes that enable them to survive (when they do survive) both the dramatic changes to which they are not already adapted, and the unintended consequences of their own functioning. Jacobs argues that there are four basic processes for evading collapse: bifurcations, positive feedback, negative feedback, and emergency adaptations. I will forego the rich details of her discussion of these processes, offering just one illustration, and then the fundamental lessons that she draws.

"Positive feedback loops" Jacobs writes "permit biomass expansion and economic expansion without loss of dynamic stability" (2000: 94). She illustrates the role that positive feedback can play through an analysis of the dynamics of Grand Banks cod fishing. This is a clear instance of intertwined natural biological and human economic systems. Processes of development and expansion led to technological innovations in trawlers, nets, sonar detection, etc., as well as growth in the yield of cod. However, the numbers of fish caught and their sizes eventually began to decline. This led a rise in prices. Rising prices, declines in fish size and yield were forms of positive feedback information. They indicated that the intelligent response would be to cut back on the rate of fishing. But in fact just the opposite happened. Instead, the fishing industry responded with more sophisticated equipment and more intensive fishing, requiring greater financial investments. But catches continued to decline, and returns on investments were disappointing: more positive feedback. Yet in 1992 Grand Banks cod fishing collapsed, "a horrendous economic and social disaster . . . to say nothing of an ecological disaster, whose ramifications are still unknown" (2000: 97). The problem, Jacobs argues, was not "the feedback information, but the response to it." The root problem, she claims, was government subsidies for the escalating growth of fishing. Had the subsidies been added into the costs, "cod would have priced itself out of the market before fish stocks collapsed" (2000: 100).

From this and similar illustrations Jacobs draws two serious lessons. First, each of these manners of evading collapse has an associated "trap." She analyzes the forms of these traps, often through the sorry lessons of human ventures that failed to beware of them (instances of what Lonergan terms "general bias").

Second, the existence of traps raises for her the question: "whether our species has inborn traits that restrain habitat destruction"? (2000: 125). Even though Jacobs is not a theologian, her question is certainly one way of posing the question of redemption. She considers several possible traits that could overcome traps leading to destruction—among them aesthetic appreciation, fear, awe. Ultimately, she concludes, it will be our intelligence, our moral consciousness, and our awareness that we *partake* of "processes of development and diversification" that we receive as *gifts* that can save us. Full awareness of being gifted is course is one way of talking about the awareness of grace (2000: 130-32, 146). Elsewhere she also identifies love as the source of recovery from collapse—as when people choose to stay and work together in declining neighbourhoods out of love of their neighbours (1993: 279-83) and when people act out of care for future generations they will never see. Like Lonergan, therefore, Jacobs is keenly aware of the need for the healing and restorative power of grace and love to reverse the effects of bias against understanding.[11]

Conclusion

The preceding sections of this article have drawn attention to the strong parallels between the works of Lonergan and Jacobs. Still, there are differences between their styles and objectives—differences that constitute complementary contributions to thinking about the interfaces between natural, human, and theological realms.

Lonergan's interests were philosophical and theological. Philosophically, he was concerned to develop a general view of the structure of the evolving natural universe and human history. His philosophy reconciles long-standing impasses between matter and spirit, nature and humanity, pragmatism and ethics. His theological objective was to offer a framework that would facilitate intelligent, critical, and compassionate theological responses to the challenges posed by ever changing natural environments and human situations. Although he made major contributions to the resolution of abstract, systematic theological problems, ultimately his concern was to enable theology to make practical contributions to the healing and transformation of the world.

Lonergan therefore dealt with issues in a generic fashion, meeting head on the intellectual difficulties that would bar philosophical and theological considerations from making their proper contributions to courses of human action. Jacobs on the other hand focused her attention on concrete and specific patterns of human (and later natural) behaviour. She began her career with very specific attention to the patterns of behaviour that made for safe and neighbourly city blocks. But the fact that those patterns depended upon factors that lay beyond the small local neighbourhoods led her progressively to ever larger and more generic considerations—to city districts and cities as wholes, to the economic interactions among cities and with non-urban regions, and eventually to the relations between urban economies and natural ecologies. As such, her work offers rich concrete illustrations that make more intelligible Lonergan's consistently generic ideas. But her work relates to that of Lonergan in a much more fundamental and important way as well. Not only do Jacobs' rich examples and concrete examples illustrate Lonergan's ideas; her analytic approach itself is an embodiment of the kind of thinking and acting that Lonergan strove to make more possible. Like Jacobs, Lonergan was a foe of big plans. He argued forcefully that practical and sound ethical actions are only possible for people "on the spot" who have the tools to also take into account the broader impact of local actions. He explicitly acknowledged Jane Jacobs as a paradigm of such thinking,[12] going so far one time as to dub her "Mrs. Insight." On the other hand, Lonergan's generalized framework also complements Jacobs' analyses, bringing to light theological concerns (e.g., the transforming effects of love) that are only tacit in Jacobs' own writings. So despite their differences approaches, there are important complementarities and convergences in their work.

Of what value is all this exploration by Lonergan and Jacobs of the dynamics of biological and economic ecologies? Jacobs herself raises this question and offers an answer:

> Is there any practical value or advantage in knowing that economic development is differentiations emerging from generalities? . . . It tells us that development isn't a collection of things but rather a process that yields things. Not knowing this, governments, their development and aid agencies, the World Bank, and much of the public in general put their faith in a fallacious 'Thing Theory' of development. The Thing Theory supposes that development is the matter of possessing things such as factories, dams, schools, tractors, whatever . . . [However, things] don't mysteriously carry the process of development with them. To suppose that [they do] creates false expectations and futilities. Worse, it evades measures that might actually foster development (2000: 32).

Just as Lonergan emphasizes that emergent probability is not primarily about things but about natural and human processes and schemes, so also does Jacobs. Processes and schemes of recurrence are the "gifts" that human actions presuppose. Hence, our focus should be upon both natural and human processes, their natural dynamics of development, and their delicate and intricate interdependencies. The better we understand them, the better we can cooperate with them. Actions that ignore or taunt these dynamics are doomed to have unintended and disastrous consequences. Whether gifts of creation or gifts of grace, they only become effective in human action when they are recognized, properly understood, and willingly cooperated with.

Both Lonergan and Jacobs could be roughly characterized as advocating a modified form of the slogan, "Think globally, act locally." For both of them, "Think globally" means to think about the future. Think about the long term. Think about the universe and human history within it as heading toward the future. It means especially, Think about *how* nature and humanity are heading toward the future. Think in ways that observe and respect the dynamics of natural and human development. That means, before anything else, properly understanding those dynamics. This way of thinking and acting differs markedly from an ethic of preserving a pristine, static "balance of nature." It implies an ethic of intimate and critical understanding and critical valuing of the natural, economic, cultural, and religious gifts that we inherit. It implies an ethic that understands these gifts as points of departure for dynamic developments. It implies learning to how to distinguish genuine dynamic developments from other initiatives that are advertised as "progress," but in fact are biased paths of decline and destruction. The work of Lonergan and Jacobs invite us take on the hard work of understanding how nature and grace are already at work developmentally, and how to better cooperate with their dynamics.

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Notes

1. See for example the announcement for the Lilly Fellows National Research Conference, "Ecology, Theology and Judeo-Christian Environmental Ethics," University of Notre Dame, February 21-24, 2002; http://www.nd.edu/~ecoltheo/.
2. See Lonergan (1992: 142-144). For a detailed account of how more or less random accumulation of prior conditions can lead to "autocatalytic" schemes of recurrence, see Kauffman (1995: 47-69).
3. See Lonergan 1992: 470-76. For a detailed analysis of this argument, see Patrick H. Byrne and Edward Hogan, "Isomorphism and Metaphysics," forthcoming.
4. Lonergan works out a more technical account of development (1992: 476-507). A detailed discussion of it and its comparisons with Jacobs' treatment is beyond the scope of this paper.
5. For especially clear instances of group bias and its resulting social deterioration, see Jane Jacobs (2000: 32-33).
6. See Augustine (1961: 63).
7. Jacobs supplies abundant, detailed economic illustrations of this developmental process (1969, especially 49-84).
8. Compare Lonergan's comment that emerging schemes do "not occur [solely] within the living thing, but go beyond it into the environment" (1992: 156).
9. Compare Lonergan's comment regarding the solution to the tension between stability and development (1992: 146-47).
10. Jacobs does not refer to Prigogine's work directly, but rather to those ideas as embodied in a book by Sally Goerner (2000: 159).
11. Interestingly, love is practically the only aspect of human life that transcends the two moral syndromes she identifies (Jacobs 1992: 121).
12. See for example Lonergan (1999b: 101).

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